

# Giuseppe Orlando

## Curriculum Vitae

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Scholar: <https://scholar.google.com/citations?hl=it&user=ubXfZvMAAAAJ>

ResearchGate: [researchgate.net/profile/Giuseppe-Orlando-6](https://researchgate.net/profile/Giuseppe-Orlando-6)

### Personal information

Born in GAGLIANO DEL CAPO (LE), ITALY, 18/10/1994

Citizenship: ITALIAN

### Education - Work experience

- 01/09/2026 - Present: **Maître de Conférences (Associate Profesor)**  
*Institution:* Institut de Mathématiques de Toulouse, INSA Toulouse (Toulouse, France)  
*Department:* Department of Mathematical and Modeling Engineering
- 01/10/2023 - 31/08/2026: **Postdoctoral researcher**  
*Institution:* École polytechnique (Palaiseau, France)  
*Laboratory and group:* CMAP (Centre de Mathématiques Appliquées), HPC@Maths group  
*Funding:* OPEN NUM DEF project (PI Prof. Marc Massot, Dr. Loïc Gouarin, Dr. Teddy Pichard (Maître de Conférences))  
*Supervisor:* Prof. Marc Massot  
*Research topic:* Development of numerical methods for liquid-gas two-phase flows with multi-scale modelling.
- 01/02/2023 - 30/09/2023: **Research fellowship (post-doc)**  
*Institution:* Politecnico di Milano (Milan, Italy)  
*Laboratory:* MOX - Department of Mathematics  
*Funding:* ESCAPE-2 H2020, projet Grant Agreement No. 800897  
*Supervisor:* Prof. Luca Bonaventura  
*Research topic:* Development of discontinuous finite elements codes for atmosphere dynamics simulations in the framework of deal.II library.
- 01/11/2019 - 31/01/2023: **Phd candidate** for the programme Mathematical Models and Methods in Engineering  
*Institution:* Politecnico di Milano (Milan, Italy)  
*Laboratory:* MOX - Department of Mathematics  
*Supervisor:* Prof. Luca Bonaventura  
*Co-supervisor:* Prof. Paolo Francesco Barbante  
*Research project:* Modelling and simulations of two-phase flows including geometric variables  
*Defence:* 18/04/2023, Milan, Italy  
*Commission:*

|                            |                                  |           |
|----------------------------|----------------------------------|-----------|
| Prof. Davide Carlo Ambrosi | Politecnico di Torino            | President |
| Prof. Simone Scacchi       | Università degli Studi di Milano | Member    |
| Prof. Luca Formaggia       | Politecnico di Milano            | Member    |
- 29/11/2021 - 15/07/2022: **Visiting student at CMAP** (Centre de Mathématiques Appliquées), École polytechnique (Palaiseau, France)
- 2016 - 2019: **Master of Science Degree** in Mathematical Engineering  
*Track:* Computational Science and Engineering  
*Institution:* Politecnico di Milano (Milan, Italy)  
*Thesis:* Development of a reactive numerical solver based on the SU2 CFD code (discussion 03/10/2019)  
*Supervisor:* Prof. Paolo Francesco Barbante  
*Mark:* 110/110
- 2013 - 2016: **Bachelor of Science Degree** in Mathematical Engineering  
*Institution:* Politecnico di Milano (Milan, Italy)  
*Thesis:* Equazioni ellittiche degeneri strutturate su campi di Hörmander (discussion 22/09/2016)  
*Supervisor:* Prof. Marco Bramanti  
*Mark:* 107/110

- 2013: **Diploma**: General High School  
*Institution*: Istruzione di Istituto Superiore G. Stampacchia (Tricase (LE), Italy)  
*Mark*: 100L/100

## Training Programmes

- 2018: Introductory course on **GPU** computing for Fluid Dynamics applications  
<https://github.com/giuseppearlando/MilanoGPU2018>
- 2019: Crash course on Vehicle Aerodynamics



## Technical skills

- Programming languages
  - **C (MPI), C++, MATLAB, FreeFem++, Python**: Academic level with the development of small projects  
[https://github.com/giuseppearlando/Progetto\\_ANEDP](https://github.com/giuseppearlando/Progetto_ANEDP)  
[https://github.com/giuseppearlando/TwoPhase\\_Flow](https://github.com/giuseppearlando/TwoPhase_Flow)  
 C++ code for incompressible Navier-Stokes equations included in the official deal.II code-gallery  
 ([https://dealii.org/developer/doxygen/deal.II/code\\_gallery\\_NavierStokes\\_TRBDF2\\_DG.html](https://dealii.org/developer/doxygen/deal.II/code_gallery_NavierStokes_TRBDF2_DG.html))  
 Development of codes in the open-source framework samurai  
 ([https://github.com/hpc-maths/2025\\_09\\_two\\_fluid\\_all\\_topology](https://github.com/hpc-maths/2025_09_two_fluid_all_topology))
  - R, Mathematica: Basic level
- Other  
**Jupyter, Paraview**: Academic level

## Language skills

- Italian: Mother language
- British English: B2 Level, TOEFL iBT Certificate, Mark 86/120
- French: Written: fair, Spoken: fair

## Computer skills

- Operating systems  
 Windows, **Linux**
- Text and spreadsheets  
 Microsoft Office, **LaTeX**

## Research interests

- Main research interests: Computational Fluid Dynamics (**CFD**), Discontinuous Galerkin (**DG**) methods, Two-phase flows, low Mach flows, **IMEX** time discretization methods, **variational** modelling
- Other research interests: Monotonization techniques, Numerical Weather Predictions (**NWP**), shallow water equations (**SWE**), lava flow models, Scientific Machine Learning (**SciML**)

## Outline of research activity

My research activity has been mainly focused on the field of Computational Fluid Dynamics (**CFD**), spanning a wide range of applications, starting from my Master thesis on the development of a numerical solver for reactive flows, with a particular emphasis on approaches which are robust in the case of vanishing species.

During my PhD studies [25], I started to focus on the modelling and simulations of **two-phase flows**, which represents my actual main research topic. In particular, I took into account geometric variables, such as the interfacial area density, whose accurate computation is a prerequisite in order to obtain reliable values of phase exchange terms [9], and the interface mean curvature [20]. This allowed me to spend some months abroad and to start a collaboration with Centre de Mathématiques Appliquées (CMAP) at École polytechnique, where I have been a postdoctoral researcher before moving to INSA Toulouse as MCF (Maître de Conférences).

My first publication [14] deals with the development of an efficient numerical scheme for the incompressible Navier-Stokes equations. The proposed approach combines flexible discontinuous finite element space discretizations with efficient semi-implicit time discretization methods and employs an artificial compressibility formulation in order to overcome some well known issues of projection methods. More recently, an extension of this approach to two-phase flows has been proposed [8].

The first significant contribution to the literature is presented in [13] and represents a non straightforward extension to the compressible Navier-Stokes equations of real gases of **IMEX** approaches originally employed for ideal gases. More in detail, a suitable fixed point loop has been introduced for the general cubic equation of state, which can effectively handle experimental data. A discontinuous Galerkin (**DG**) spatial discretization has been employed, so as to easily achieve a full second-order method. Moreover, suitable refinement indicators have been introduced for non-ideal gases, so as to effectively use Adaptive Mesh Refinement (**AMR**). An analysis of the employed second order IMEX-RK method has been carried out, providing an improvement in terms of absolute monotonicity.

I also focused on the application of the IMEX scheme proposed in [13] to atmosphere dynamics and Numerical Weather Prediction (**NWP**) [11, 23], showing the effectiveness of the method in a number of test cases even with non-trivial and real orography profiles. The efficiency of the proposed method and solver have been demonstrated in [18, 19]. Moreover, for the first time in the literature, the robustness of the method using non-conforming meshes [7] and curved elements [6] for flows over orography has been proven. Finally, an analysis of a general class of IMEX-RK methods in terms of asymptotic-preserving (**AP**) property has been recently performed [5] and higher order methods with respect to that employed in [13] have been validated [4, 5].

In addition, I developed a filtering monotonization technique [12] in order to avoid (or at least reduce) the oscillations that arise when high order discretization methods are used. Its potentialities have been shown in a number of benchmarks, including test cases for the Euler equations of gas dynamics.

Furthermore, I also tested for the first time in the literature to use **language models** for the development of solvers for PDEs based on the finite element method [10].

Nowadays, I am mainly working on the modelling and simulations of **two-phase flows**, with particular emphasis on the derivation of models through **variational** principles [3, 15–17, 21] and on the analysis of a hierarchy of Baer-Nunziato type models for compressible two-phase flows [2]. Furthermore, I am working on the development of robust and accurate models for the shallow water (de Saint-Venant) equations [1], including lava flow models [24], as well as on development of high-order IMEX-RK methods for (degenerate) diffusion-reaction equations [22].

## Scientific publications

### List of publications

- [1] L. Arpaia, G. Orlando, C. Ferrarin, and L. Bonaventura. “High-order adaptive discontinuous finite elements for the shallow water equations with sub-grid irregular bathymetry”. *Advances in Water Resources* 217 (2026), p. 105472. DOI: 10.1016/j.advwatres.2026.105472.

- [2] G. Orlando, W. Haegeman, M. Pelanti, and M. Massot. “A robust computational framework for the mixture-energy-consistent six-equation two-phase model with instantaneous mechanical relaxation terms”. *Computers & Fluids* 317 (2026), p. 107193. DOI: 10.1016/j.compfluid.2026.107193.
- [3] W. Haegeman, G. Orlando, S. Kokh, and M. Massot. “An all-topology two-fluid model for two-phase flows derived through Hamilton’s Stationary Action Principle”. *Proceedings of the Royal Society A* 482 (2026), p. 20250835. DOI: 10.1098/rspa.2025.0835.
- [4] G. Orlando, S. Boscarino, and G. Russo. “A quantitative comparison of high-order asymptotic-preserving and asymptotically-accurate IMEX methods for the Euler equations with non-ideal gases”. *Computer Methods in Applied Mechanics and Engineering* 442 (2025), p. 118037. DOI: 10.1016/j.cma.2025.118037.
- [5] G. Orlando and L. Bonaventura. “Asymptotic-preserving IMEX schemes for the Euler equations of non-ideal gases”. *Journal of Computational Physics* 529 (2025), p. 113889. DOI: 10.1016/j.jcp.2025.113889.
- [6] G. Orlando, T. Benacchio, and L. Bonaventura. “Impact of curved elements for flows over orography with a Discontinuous Galerkin scheme”. *Journal of Computational Physics* 519 (2024), p. 113445. DOI: 10.1016/j.jcp.2024.113445.
- [7] G. Orlando, T. Benacchio, and L. Bonaventura. “Robust and accurate simulations of flows over orography using non-conforming meshes”. *Quarterly Journal of the Royal Meteorological Society* 150 (2024), pp. 4750–4770. DOI: 10.1002/qj.4839.
- [8] G. Orlando. “An implicit DG solver for incompressible two-phase flows with an artificial compressibility formulation”. *International Journal for Numerical Methods in Fluids* 96 (2024), pp. 1932–1959. DOI: 10.1002/flid.5328. Preprint: arXiv: 2307.04580.
- [9] G. Orlando, P.F. Barbante, and L. Bonaventura. “On the evolution equations of interfacial variables in two-phase flows”. *International Journal of Multiphase Flow* 169 (2023), p. 104617. DOI: 10.1016/j.ijmultiphaseflow.2023.104617.
- [10] G. Orlando. “Assessing ChatGPT for coding finite element methods”. *Journal of Machine Learning for Modeling and Computing* (2023), pp. 135–171. DOI: 10.1615/JMachLearnModelComput.2023049326.
- [11] G. Orlando, T. Benacchio, and L. Bonaventura. “An IMEX-DG solver for atmospheric dynamics simulations with adaptive mesh refinement”. *Journal of Computational and Applied Mathematics* (2023), p. 115124. DOI: 10.1016/j.cam.2023.115124.
- [12] G. Orlando. “A filtering monotone approach for DG discretizations of hyperbolic problems”. *Computers & Mathematics with Applications* 129 (2023), pp. 113–125. DOI: 10.1016/j.camwa.2022.11.017. Preprint: arXiv: 2204.08693.
- [13] G. Orlando, P.F. Barbante, and L. Bonaventura. “An efficient IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases”. *Journal of Computational Physics* 471 (2022), p. 111653. DOI: 10.1016/j.jcp.2022.111653.
- [14] G. Orlando et al. “An efficient and accurate implicit DG solver for the incompressible Navier-Stokes equations”. *International Journal for Numerical Methods in Fluids* 94 (2022), pp. 1484–1516. DOI: 10.1002/flid.5098.

## List of proceedings

- [15] W. Haegeman, G. Orlando, S. Kokh, and M. Massot. “Modelling of relative velocity, velocity fluctuations and their interactions for two-fluid models by Stationary Action Principle”. *ESAIM: Proceedings and Surveys* (2026). (in press). Preprint: hal-05224712.
- [16] W. Haegeman et al. “A Two-Phase Flow Model with Scale Separation for the Separated-to-Disperse Phase Transition”. *Advances in Entry Systems Modeling: Modeling Summer Visit 2025 Technical Reports*. Ed. by Bruno Dias et al. Vol. TM-20260002490. NASA Technical Memorandum. <https://ntrs.nasa.gov/citations/20260002490>. NASA Ames Research Center, 2026, pp. 38–42. DOI: 10.64631/EPFF4875.

- [17] W. Haegeman, S. Kokh, M. Massot, and G. Orlando. "A generic framework to derive systems of conservation laws with source terms and its application to heat conduction in fluid flows: An alternative to the method of moments in kinetic theory of gases?" *ESAIM: Proceedings and Surveys* 78 (2025), pp. 80–97. DOI: 10.1051/proc/202578080.
- [18] G. Orlando, T. Benacchio, and L. Bonaventura. "Improving the scalability of a high-order atmospheric dynamics solver based on the deal.II library". *Procedia Computer Science* 267 (2025), pp. 227–236. DOI: 10.1016/j.procs.2025.08.249. Proceedings of the Third EuroHPC Day.
- [19] G. Orlando, T. Benacchio, and L. Bonaventura. "Efficient and scalable atmospheric dynamics simulations using non-conforming meshes". *Procedia Computer Science* 255 (2025), pp. 33–42. DOI: 10.1016/j.procs.2025.02.258. Proceedings of the Second EuroHPC Day.
- [20] G. Orlando, P. F. Barbante, and L. Bonaventura. "Assessment of Different Strategies to Compute Surface Tension Contribution". *Numerical Mathematics and Advanced Applications ENUMATH 2023, Volume 2: European Conference, September 4-8, Lisbon, Portugal*. Vol. 154. Springer Nature. 2025, pp. 213–222. DOI: 10.1007/978-3-031-86169-7\_22. Preprint: arXiv: 2402.04670.

## List of preprints

- [21] W. Haegeman, G. Orlando, S. Kokh, and M. Massot. *A two-scale two-phase flow model for the separate-to-disperse phase transition in atomizing flows*. 2026. URL: <https://hal.science/hal-05673253>.
- [22] P. F. Antonietti, M. Corti, and G. Orlando. *Optimized high-order IMEX-RK schemes for degenerate diffusion-reaction problems with application to travelling waves phenomena*. 2026. arXiv: 2606.15726 [math.NA].
- [23] L. Bottani et al. *An IMEX-DG solver with non-conforming mesh refinement for atmospheric dynamics with rotation*. 2026. arXiv: 2605.29598 [math.NA].
- [24] F. Gatti and G. Orlando. *Second-order optimally stable IMEX (pseudo-)staggered Galerkin discretization with application to depth-integrated lava flow simulations*. 2026. arXiv: 2509.09460 [math.NA].

## PhD thesis

- [25] G. Orlando. "Modelling and simulations of two-phase flows including geometric variables". <http://hdl.handle.net/10589/198599>. PhD thesis. Politecnico di Milano, 2023.

## Master thesis

- [26] G. Orlando. *Development of a reactive numerical solver based on the SU2 CFD code*. <http://hdl.handle.net/10589/150021>. Master thesis, Politecnico di Milano. 2019.

## Editorial activity

- 01/01/2026 - :Associate editor for *Monthly Weather Review (MWR)*
- I have acted as a referee for international journals in Mathematical Modelling and Scientific Computing: *Computers and Mathematics with Applications (CAMWA)*, *Journal of Computational Physics (JCP)*, *Computer & Fluids (CAF)*, *Monthly Weather Review (MWR)*, *International Journal on Geomathematics (GEM)*.

## Main conferences participation

- 12th International Conference on Numerical Methods for Multi-Material Fluid Flow (MultiMat 2026)  
*Date and Location:* 11/09/2026, Biarritz (France)  
*Title and Duration:* "A two-scale two-phase model for the separated-to-disperse phase transition by atomization", 30 minutes
- **Invited talk** in the minisymposium "Advances in Computational Models and Methods for Multiphase Flows with Phase Change" at 17th World Congress on Computational Mechanics – 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM – ECCOMAS 2026)  
*Organizers:* L. Brandt, S. Jain, M. Pelanti, M. Rodriguez, and A. Roccon  
*Date and Location:* 21/07/2026, Munich (Germany)

*Title and Duration:* "A unified compressible two-phase, two-scale model with inter-scale mass transfer for separated-to-disperse transitions in atomizing flows", 15 minutes

- **Invited seminar** for Workshop Comfluans  
*Date and Location:* 07/07/2026, IMAG, Montpellier (France)  
*Title and Duration:* "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows", 45 minutes
- 4th International workshop on Perspectives on Multiphase Fluid Dynamics, Continuum Mechanics and Hyperbolic Balance Laws (ProHyp2026)  
*Date and Location:* 02/06/2026, Strasbourg (France)  
*Title and Duration:* "An *all-topology* two-fluid model for two-phase flows derived from Hamilton's Stationary Action Principle", 30 minutes
- **Invited seminar** for the series "Séminaire Analyse Appliquée (AA)" at I2M (Institut de Mathématiques de Marseille)  
*Date and Location:* 31/03/2026, Marseille (France)  
*Title and Duration:* "An *all-topology* two-fluid model for two-phase flows derived through Hamilton's Stationary Action Principle", 45 minutes
- Speaker Flash workshop  
*Date and Location:* 02/02/2026, Palaiseau (France)  
*Title and Duration:* "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows", 50 minutes
- **Invited speaker** at 38ème séminaire de mécanique des fluides numérique  
*Date and Location:* 26/01/2026, Paris (France)  
*Title and Duration:* "A unified compressible two-phase, two-scale model with capillarity for separated-to-disperse transitions in atomizing flows", 40 minutes
- **Invited seminar** for the series "Séminaires de Mathématiques et de leurs Applications" at LMAP (Laboratoire de Mathématiques et de leurs Applications, Pau)  
*Date and Location:* 22/01/2026, Pau (France)  
*Title and Duration:* "An adaptive high-order DG solver for the shallow water equations with irregular bathymetry", 45 minutes
- **Invited speaker** at ARIA (Accurate ROMs for Industrial Applications) 2025 workshop  
*Date and Location:* 23/09/2025, Biarritz (France)  
*Title and Duration:* "A high-order matrix-free adaptive DG solver for the shallow water equations with irregular bathymetry", 30 minutes
- **Invited talk** in the minisymposium "High-Order and Innovative Methods for Coupled Problems in Life Science and Geophysics" at 11th International Conference on Coupled Problems in Science and Engineering (CO'UPLED 2025)  
*Organizers:* F. Bonizzoni, M. Corti, and S. Pagani  
*Date and Location:* 27/05/2025, Villasimius (Italy)  
*Title and Duration:* "A high-order solver for NWP using non-conforming adaptive and curved meshes: applications to flows over orography", 15 minutes
- 12th International Conference on Multiphase Flow (ICMF 2025)  
*Date and Location:* 14/05/2025, Toulouse (France)  
*Title and Duration:* "A unified two-scale gas-liquid two-fluid model with capillarity and interface regularization through a mass transfer between scales", 15 minutes
- **Invited seminar** for the series "Séminaire de Calcul Scientifique et Modélisation" at IMB (Institut de Mathématiques de Bordeaux)  
*Date and Location:* 10/04/2025, Bordeaux (France)  
*Title and Duration:* "Asymptotic-preserving IMEX schemes for the Euler equations of non-ideal gases", 45 minutes
- **Invited speaker** at Seminar Mathematics and Atmospheric Physics  
*Date:* 23/10/2024

*Title and Duration:* "Robust and accurate simulations of flows over orography using non-conforming variable resolution meshes", 45 minutes

- Seminar HPC@Maths  
*Date and Location:* 18/10/2024, Palaiseau (France)  
*Title:* "Numerical analysis of a two-fluid model"
- European Meteorological Society (EMS) Annual Meeting  
*Date and Location:* 06/09/2024, Barcelona (Spain)  
*Title and Duration:* "Robust and accurate simulations of flows over orography using non-conforming variable resolution meshes", 15 minutes
- 2nd BICTAM-CISM Symposium on Dispersed Multiphase Flows: from Measuring to Modeling  
*Date and Location:* 22/08/2024, Beijing (China)  
*Title and Duration:* "A unified two-scale gas-liquid two-fluid model with capillarity and interface regularization through a mass transfer between scales", 15 minutes
- International Conference on Numerical Methods in Multiphase Flows - 5 (ICNMMF-5)  
*Date and Location:* 28/06/2024, Reykjavik (Iceland)  
*Title and Duration:* "A two-scale model for two-phase flows including mass transfer", 15 minutes
- **Invited talk** in the minisymposium "Computational Methods for Multiphase Flows with Liquid-Vapor Transition" at 9th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS 2024)  
*Organizers:* L. Brandt, M. Pelanti, M. Rodriguez, and S. Tanguy  
*Date and Location:* 06/06/2024, Lisbon (Portugal)  
*Title and Duration:* "A two-scale model for two-phase flows including geometric variables and mass transfer", 15 minutes
- 46ème Congrès National d'Analyse Numérique (CANUM 2024)  
*Date and Location:* 29/05/2024, Ile de Ré (France)  
*Title and Duration:* "A two-scale model for two-phase flows including geometric variables and mass transfer"; 15 minutes
- **Invited speaker** at the INdAM Workshop INSiDEs (Innovations in the Numerical Treatment of Stiff Differential Equations)  
*Date and Location:* 22/02/2024, Rome (Italy)  
*Title and Duration:* "An asymptotic-preserving scheme for Euler equations with non-ideal gases", 45 minutes
- Seminar HPC@Maths  
*Date and Location:* 24/11/2023, Palaiseau (France)  
*Title:* "Derivation of relations for interfacial area density in two-phase flows through the Stationary Action Principle"
- **Invited talk** in the minisymposium "Theoretical and computational aspects of discontinuous Galerkin methods" at European Conference on Numerical Mathematics and Advanced Applications (ENUMATH 2023)  
*Organizers:* V. Aizinger and A. Rupp  
*Date and Location:* 05/09/2023, Lisbon (Portugal)  
*Title and Duration:* "A filtering monotoneization technique for DG discretization of hyperbolic problems", 25 minutes
- Bi-annual congress of SIMAI 2023  
*Date and Location:* 29/08/2023, Matera (Italy)  
*Title and Duration:* "A filtering monotoneization technique for DG discretization of hyperbolic problems", 25 minutes
- **Invited talk** in the minisymposium "Computational Advances in Earth System Modeling" at 17th U. S. National Congress on Computational Mechanics (USNCCM17)  
*Organizers:* M. Perego, A. Salinger, and I. Tezaur  
*Date and Location:* 25/07/2023, Albuquerque (USA)  
*Title and Duration:* "An IMEX-DG solver for atmospheric dynamics with Adaptive Mesh Refinement", 20 minutes
- PDEs on the sphere 2023 (PDEs 2023)  
*Date and Location:* 06/07/2023, Grenoble (France)  
*Title and duration:* "An IMEX-DG solver for atmospheric dynamics based on deal.II framework", 30 minutes

- 7th ECCOMAS Young Investigators Conference (YIC2023)  
*Date and Location:* 21/06/2023, Porto (Portugal)  
*Title and Duration:* "An IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases", 15 minutes
- **Invited seminar** for the series "Seminario di Modellistica Differenziale Numerica" at Sapienza Università di Roma  
*Date and Location:* 22/11/2022, Rome (Italy)  
*Title and Duration:* "A filtering monotonicization technique for DG discretizations of hyperbolic problems", 45 minutes
- 14th European Fluid Mechanics Conference (EFMC14)  
*Date and Location:* 14/09/2022, Athens (Greece)  
*Title and Duration:* "An IMEX-DG solver for the compressible Navier-Stokes equations for non-ideal gases", 12 minutes
- XVIII International Conference on Hyperbolic Problems: Theory, Numerics, Applications (HYP2022)  
*Date and Location:* 21/06/2022, Malaga (Spain)  
*Title and Duration:* "An IMEX-DG solver for the compressible Navier-Stokes equations with a general equation of state", 15 minutes
- 45ème Congrès National d'Analyse Numérique (CANUM 2020)  
*Date and Location:* 16/06/2022, Evian-les-Bains (France)  
*Title and Duration:* "An IMEX-DG solver for the compressible Navier-Stokes equations with a general equation of state", 15 minutes
- Bi-annual congress of SIMAI 2020+2021  
*Date and Location:* 30/08/2021, Parma (Italy)  
*Title and Duration:* "An accurate and efficient implicit Discontinuous Galerkin solver for the incompressible Navier-Stokes equations", 25 minutes
- Conference "Numerical methods for hyperbolic problems (NumHyp 2021)"  
*Date:* 26/07/2021  
*Title and Duration:* "An efficient and accurate implicit DG solver for the incompressible Navier-Stokes equations", 30 minutes

## Outline of teaching activity

### Responsible teacher

- Crash course on the use of MATLAB software for students - Spring Semester - A.Y. 2019-2020 and 2020-2021  
*Teaching load:* 16 hours (per course)  
*Institution:* Politecnico di Milano - Lecco Campus (Italy)

### Exercise Session

- APM51656EP “Elements of measure theory and functional analysis” - A.Y. 2025-2026  
*Teaching load:* 6 hours  
*Institution:* École polytechnique (France)  
*Path:* Master Applied Mathematics and Statistics (M1)  
*Responsible of the course:* Prof. David Gérard-Varet
- MECCANICA RAZIONALE - A.Y. 2020-2021  
*Teaching load:* 20 hours  
*Institution:* Politecnico di Milano (Italy)  
*Path:* Bachelor degree in Mathematical Engineering (Italian)  
*Responsible of the course:* Prof. Maurizio Stefano Vianello
- MATHEMATICAL MODELS AND METHODS IN ENGINEERING [1] - A.Y. 2019-2020 and 2020-2021  
*Teaching load:* 10 hours (A.Y. 2019-2020), 20 hours (A.Y. 2020-2021)  
*Institution:* Politecnico di Milano (Italy)  
*Path:* Master degree in Mathematical Engineering  
*Responsible of the course:* Prof. Maurizio Stefano Vianello

### Informatic Laboratory

- **Informatic Laboratory** for the course SHAPES AND ALGORITHMS: TOOLS FOR GENERATIVE DESIGN - A.Y. 2020-2021  
*Teaching load:* 26 hours  
*Institution:* Politecnico di Milano (Italy) *Responsible of the course:* Prof. Anna Scotti

### Tutoring

- **Tutor of the final project** for the APM51051EP course “Systèmes dynamiques pour la modélisation et la simulation des milieux réactifs multi-échelles” (M2) - A.Y. 2024-2025, 2025-2026  
*Institution:* École polytechnique (France)  
*Responsible of the course:* Prof. Marc Massot
- **Tutor of the final project** for the MAP551 course “Systèmes dynamiques pour la modélisation et la simulation des milieux réactifs multi-échelles” (M2) - A.Y. 2023-2024  
*Institution:* École polytechnique (France)  
*Responsible of the course:* Prof. Marc Massot

## Supervision activity Master thesis

1. Safira Leandra Liora Piasko (September 2022 - March 2023)  
*Institution:* ETH Zurich (Switzerland)  
*Title:* Discontinuous Finite Element Methods for Large Eddy Simulation (LES) of Incompressible Flows  
*Collaborators:* Prof. Luca Bonaventura, Prof. Antonella Abbà, Prof. Christoph Schär
2. Letizia Bottani (September 2025 - February 2026)  
*Institution:* Double program Politecnico di Milano (Italy) - Technical University of Denmark (Denmark) *Title:* Simulation of rotating flows with a high-order finite element framework for atmospheric dynamics  
*Collaborators:* Prof. Luca Bonaventura, Dr. Tommaso Benacchio, Prof. Allan Peter Engsig-Karup

## Projects, organizations and other info

- 17/11/2025: **Co-organiser** of the practical training session "Hands-on Samurai", dedicated to introducing the samurai code to approximately 25 external participants from 10 different research institutions (held on site at CMAP)

- 07/07/2025 - 01/08/2025 : Participation to Modeling Summer Visit (MSV) Program as co-PI of a research project.  
*Title:* A two-phase flow model with scale separation for the separated-to-disperse phase transition  
*Institution:* NASA Ames  
*Funding:* Partial support by Fondation Mathématique Jacques Hadamard (2k €) through the program Junior Scientific Visibility
- **Co-organization of seminars HPC@Maths** at CMAP - École polytechnique starting from November 2023
- **Co-organisation of a minisymposium** at SIMAI 2025 Congress  
*Title:* Recent advancements in numerical methods and physical modelling of multi-material flows  
*Co-organizers:* Niccolò Tonicello (SISSA), Federico Dalla Barba (Padua University), Marica Pelanti (ENSTA)
- 28/08/2023 - 01/09/2023: **Staff member** of Bi-annual congress of SIMAI 2023
- Solver for incompressible Navier-Stokes equations included in the **official deal.II code-gallery**  
[https://dealii.org/developer/doxygen/deal.II/code\\_gallery\\_NavierStokes\\_TRBDF2\\_DG.html](https://dealii.org/developer/doxygen/deal.II/code_gallery_NavierStokes_TRBDF2_DG.html)
- PI of the EuroHPC Benchmark project EHPC-BEN-2024B03-045  
*Title:* A high-order efficient and scalable solver for atmospheric dynamics using non-conforming and curved meshes
- PI of the EuroHPC Development project EHPC-DEV-2024D10-054  
*Title:* A high-order efficient and scalable solver for NWP using dynamically adaptive non-conforming meshes
- PI of the ISCRA-C Project FEM-GPU  
*Title:* Finite Element Methods for GPU architectures
- PI of the ISCRA-C Project NUMNETF  
*Title:* NUMerical Methods for Non-Equilibrium Two-phase Flows  
*Research Group:* Paolo Francesco Barbante, Luca Bonaventura  
*Amount:* 1 720 €(=86k core hours on G100 cluster, CINECA, Italy)
- PI of the ISCRA-C Project SIDICONS  
*Title:* Semi-Implicit Discretizations for Incompressible and COmpressible Navier-Stokes equations  
*Research group:* Antonella Abbà, Paolo Francesco Barbante, Luca Bonaventura

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